

Species Differences in Executive Function Correlate With Hippocampus Volume and Neocortex Ratio Across Nonhuman Primates

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A persistent debate in behavioral research is whether brain size or architecture relates to cognitive performance. A growing body of evidence has demonstrated correlations between brain size and ecological and behavioral tasks. These studies are premised on a causal link between brain size and cognitive function, although this association has little empirical backing. We show, for a set of 46 species from 17 primate genera, that competence on a series of eight executive function cognitive tasks both correlate across tasks and with brain size and architecture across species. Our model selection approach showed that, although several measures of brain component volumes are significantly associated with performance, hippocampus size is the best predictor of overall performance. The best performing model also includes total brain size and relative neocortex size. Additionally, absolute measures are much more predictive of performance than relative measures of brain and brain component size. These results are consistent with the hippocampus' role in learning, and the executive brain (neocortex) being important for problem solving and consolidation. Our findings challenge and extend those of previous analyses by clarifying the relationship between overall brain size and specific regional volumes. They also suggest that commonly used indices of encephalization, such as residuals of brain volume regressed on body size, may confound rather than clarify matters.

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Jerison's (1973) seminal analyses of brain evolution and intelligence in vertebrates postulate that large brains must confer cognitive benefits in order for animals to bear the costs of growing and maintaining them. However, the evidence for whether cognitive abilities are related to brain size (or any component of brain volume) remains, at best, scarce. Even more profoundly, we have little evidence that cognitive abilities even vary significantly between species. Indeed, Preuss (2001) went as far as to say that encephalization is "uninterpretable in terms of cognition or behavioral processes." Whether this claim derives from a lack of evidence or of an absence of variation in cognitive performance is unclear. Furthermore, Macphail (1985) famously argued that, given the evidence available, "we should . . . conclude that there are no differences [in intelligence], either qualitative or quantitative, among vertebrates (excluding man)." The basis of MacPhail's argument is not necessarily that all species have equal cognitive ability, but rather that the context of a cognitive task can strongly influence motivation and perception. He, therefore, argues that we should not expect to compare species performance on standardized

tasks quantitatively or qualitatively. While Macphail's position has always been justified (it is, in effect, the null hypothesis), it has rested uneasily on the question of why species should differ in brain size if they do not differ in cognitive abilities. This variation requires an explanation, given that the brain is one of the energetically most expensive tissues both to develop and to maintain (Aiello & Wheeler, 1995; Karbowski, 2007). However, it is possible that McPhail's scepticism may simply reflect the fact that, so far, few attempts have been made to compare performance across species so as to demonstrate categorically whether or not a consistent quantitative difference in ability exists.

This is an important issue because, irrespective of any differences that may exist in brain organization between species, the raw fact is that we still have to explain why there are such massive volumetric differences in brain size (Dunbar & Shultz 2007a). Additionally a large, and growing, body of literature has identified a number of behavioral and ecological characteristics that are associated with large brain size across vertebrates. These studies are premised on an underlying causal link between brain size and cognitive function (e.g., Lefebvre & Sol, 2008). For example, in primates, brain size is associated with the size of both social groups (Dunbar, 1992; Barton, 1996) and grooming cliques (Kudo & Dunbar, 2001), deception rates (Byrne & Corp, 2004), coalition formation (Dunbar & Shultz 2007b), and social learning and tool use (Reader & Laland, 2002). In birds, brain size is correlated with both behavioral traits (learning and innovation: Lefebvre, Reader, & Sol, 2004; tool use: Lefebvre, Nicolakakis & Boire, 2002) as well as ecological ones (population trends: Shultz, Bradbury, Evans, Gregory, and Blackburn, 2005; invasion success: Sol, Duncan, Black-

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burn, Cassey, and Lefebvre, 2005; and survival: Sol, Szekely, Liker, & Lefebvre, 2007).

However tantalizing these relationships are as adaptive explanations for encephalization, the cognitive requirements for behavioral traits such as sociality or ecological flexibility remain speculative (Healey & Rowe, 2007). Without a foundation that underpins the link between brain architecture and cognition, these papers amount to little more than adaptive storytelling. Thus, to validate the claim that large brains are associated with 'intelligence,' it is essential to demonstrate that species with larger brains perform better on cognitive tasks. Recently, cognitive performance on conventional tasks have been collated to demonstrate species differences in "general intelligence" (Johnson, Deaner, & van Schaik, 2002), with some qualitative evidence to suggest that this might relate to differences in brain size (Lee, 2006; Deaner, Isler, Burkart, & van Schaik, 2007). However, while these studies are indicative, both the performance and the brain size measurements they use are fairly crude. The data available on actual performance (rather than relative performance ranked across studies, the index used by Deaner et al., 2007) may provide more insight into how brain size (and brain architecture) relates to ability to learn specific tasks.

A second issue of considerable evolutionary significance is whether any such differences in cognition are a function of differences in brain size as a whole, or whether evaluating components of the brain associated with learning and memory may be more informative. Proponents of domain general cognition argue that the brain is an interconnected network of pathways and that total brain size is potentially the best measure of cognitive potential (e.g., Lefebvre & Sol, 2008). In contrast, controlled cognitive experiments have demonstrated that, in some cases at least, brain components accurately predict performance. One key region is the hippocampus, which is widely implicated in learning and memory formation. Lesions to the hippocampus system result in anterograde amnesia, spatial disorientation, and associative learning deficits. Implicit learning, however, seems to be relatively unaffected. Therefore, the hippocampus appears to be critical for consolidation of memories, especially of episodic or executive functioning information (Brasted, Bussey, Murray, & Wise, 2003; McClelland, McNaughton, & O'Reilly, 1995).

O'Keefe and Nadel (1978) suggested the major role of the hippocampus was for encoding and retrieving spatial memory. This has led to an emphasis in animal literature on spatial tasks. For example, hippocampus volume is associated with spatial memory and learning in birds (e.g., Healy & Krebs, 1992; Krebs, Sherry, Healy, Perry, and Vaccarino, 1989; Sherry, Jacobs, & Gaulin, 1992). As hippocampus size is temporally variable (e.g., Smulders, Sasson, & DeVogd, 1995), and is likely to be influenced by experience (e.g., taxi drivers in London: Maguire et al., 2000; see also Clayton & Krebs, 1994), using gross morphological characteristics has been criticized on the grounds that it masks inherent variation in brain architecture (Healy & Rowe, 2007). Nonetheless, absolute hippocampal volume does vary across species, and regardless of intraspecific variation hippocampal volume should be expected to correlate with learning tasks if size and performance have a direct relationship.

In addition to the hippocampus, the neocortex (in particular, the frontal and medial temporal lobes) is also critical for learning and problem solving. More specifically, the frontal lobe (and especially the prefrontal cortex) has been implicated as the "center" for execu-

tive function (Miller & Cohen, 2001; Thompson et al., 2001). The executive function system goes beyond primary perceptual processing and automatic decision-making by requiring inferences to be made from primary inputs; it is thought to comprise a suite of individual cognitive abilities such as planning, memory, inhibition, and inference that are involved in ordering and organizing cognitive processes (Stuss, Eskes, & Foster, 1994). It has also been suggested that executive function competences may underpin advanced social-cognitive skills such as theory of mind that may be unique to humans (Ozonoff, 1995; Mitchell, 1997). Although the neocortex (or isocortex) is an integral component of the brains of all amniotes, large neocortices are a primate speciality (Finlay & Darlington, 1995). The alternative hypothesis is that the brain operates in a more distributed fashion (as has alternatively been argued to be the case, e.g., with theory of mind itself: Gallagher & Frith, 2003). In this case, we might expect total brain size or neocortex volume to correlate better with cognitive ability than the frontal lobe or other subcortical brain regions. By evaluating studies of comparative cognition, we may be able to bridge the gap between gross comparative studies relating brain size to general characteristics and more experimental approaches that seek to identify relationships between brain components and learning.

We test these claims quantitatively by asking whether there are consistent quantitative differences across species in performance on executive function tasks that correlate with components of brain volume. We have expanded the approach used by Deaner and colleagues (Deaner et al., 2007; Deaner, van Schaik, & Johnson, 2006; Johnson et al., 2002), by compiling performance on executive functioning tasks. Our approach refines the previous studies in three important areas: 1) we include raw performance scores on tasks rather than aggregating performance into a general intelligence score, 2) we include additional brain components associated with learning in our models, and 3) we build and compare performance of multivariate models rather than univariate ones. We test three explicit hypotheses:

- 1) As the hippocampus is central to learning and memory acquisition (Brasted et al., 2003), cognitive performance should be positively correlated with hippocampal volume.
- 2) As the neocortex is the center of executive function tasks, the total (or relative) size of the neocortex should be positively correlated with cognitive performance.
- 3) If learning is a domain general process, with multiple brain areas implicated in processing, ordering, and interpreting information, cognitive performance should be correlated with total brain size.

Finally, as hypotheses 1–3 are not mutually exclusive, we use a model selection process that includes multiple brain measures since a multivariate model may provide the best predictive model of cognitive performance.

Method

We examined a comprehensive corpus of experimental tests on primates (Johnson et al., 2002) and selected from these a set of papers that provided comparative data across a range of species on eight core executive function tasks subject to two criteria: (1) that

the same design was used with each species, and (2) that the outcome variable was measured as a percentage of correct trials in the test phase (so as to allow us to compare performance by different species directly). We also added a number of more recent papers that reported studies of additional species on these tasks. The tasks were as follows:

(a) The **reverse reward task** is a test of inhibition. The subject is shown two differing quantities of food; then, whichever it selects, it receives the opposite. In order to be successful, the subject has to be able to inhibit its first response (i.e., selecting the larger quantity of food).

(b) The **short term memory test** (delay) is a simple memory task, where food is hidden under one of two cups, in sight of the subject, who then has to remember the food's location for 30 seconds before being allowed to make a selection.

(c) The **reversal learning task** is a well-known paradigm, where one of two stimuli is rewarded consistently until the subject reaches criterion, at which point the reward rule is reversed. For present purposes, the dependent variable is the percentage of correct selections in the set of trials after reversal.

(d) The **concept transference** (or learning set) **task** tests animals' ability to generalize a learned concept. Subjects are first taught a simple oddity paradigm until they reach criterion, after which they are given 100 trials using sets of three novel stimuli.

(e) The **oddy task** uses two identical objects and a third different object as the stimuli; the correct response was to choose the odd object.

(f) The **patterned string problem** presents individuals with multiple strings, of which only one is attached to a reward.

(g) The **invisible displacement task** tests an individuals' understanding of object permanence by hiding an object under an inverted cup and then transferring it to beneath an adjacent cup, using a displacer to obscure the object from view when doing so.

(h) Finally, the **detour task** involves threading a bent wire with a reward; individuals were given 45 seconds to figure out how to remove the reward from the bent wire.

These tasks can be roughly grouped into discrimination (oddy, concept transference), spatial (detour, string, invisible displace-

ment), memory (delay), and inhibition (reverse reward, reversal problems). In all of these paradigms, the data are presented as the percentages of correct trials, which are not appropriate for parametric statistics. Since an arcsin transformation is the most appropriate when data are in the form of proportions, we converted the percent correct responses to proportions and then arcsin-transformed these (hereafter ability).

Performance data were collated for 122 task-by-species combinations, involving a minimum of 444 individual animals from 46 species (for sample sizes and original sources: see *Supplementary Online Material*). For some species, multiple studies were available for a single task. As data were aggregated over numerous studies ($N = 43$), we first evaluated the effect of biases introduced by the inclusion of different studies by running a General Linear Model (GLM) with ability as the response, genus and task as factors and study as a random factor. We incorporated two interactions: genus*task and genus*task*study (the three-way interaction will identify species differences in performance within tasks across studies). There were significant differences across tasks ($F_{9,28.48} = 13.19, p < .001$) and genus ($F_{20,29.48} = 7.34, p < .001$). There was a significant interaction between task*genus ($F_{53,28.19} = 2.85, p = .002$), indicating that relative genera performance varied across tasks. However, there was no significant interaction between genus*task*study ($F_{25,20} = 1.26, p = .30$). We, therefore, excluded study as a variable in all further analyses, and averaged duplicated genus' performances such that there was a single measure for each genus for each task. A summary of which genera were included for each task is shown in Table 1. Full information on individual studies is available in the supplementary information (Table S1).

Brain Measures

Absolute neocortex volume and total brain volume data were obtained from Stephan, Frahm, and Baron (1981), based on histological samples. Our primary analyses used absolute volumes of brain, body, neocortex, subcortical brain volume (i.e., total brain minus neocortex), visual (i.e., V1) and non-V1 neocortex (i.e.,

Table 1
Genera Representation Across Different Tasks

Genus	Reverse reward	Delay	Reversal	String	Learning sets	Oddity	Displacement	Detour
<i>Ateles</i>			X	X	X			
<i>Callithrix</i>			X		X			
<i>Cebus</i>		X	X	X	X	X	X	X
<i>Cercocebus</i>	X	X		X	X			
<i>Cercopithecus</i>		X		X		X		X
<i>Eulemur</i>	X				X			
<i>Gorilla</i>	X		X	X	X	X	X	
<i>Lagothrix</i>		X				X		X
<i>Lemur</i>		X	X		X	X		X
<i>Macaca</i>	X	X	X	X	X	X	X	X
<i>Mandrillus</i>		X		X				
<i>Miopithecus</i>			X					
<i>Nycticebus</i>					X			
<i>Pan</i>	X	X	X	X	X	X	X	X
<i>Saguinus</i>	X						X	
<i>Saimiri</i>	X		X		X	X	X	X
<i>Varecia</i>		X						

neocortex minus V1), medulla, hippocampus, and cerebellum, along with neocortex ratio (neocortex/rest of brain). We have included the medulla and cerebellum as parts of the brain that are not explicitly associated with associative learning. Since we would expect the relationship with these components to be less robust than with areas of the brain that are associated with learning (e.g., neocortex and hippocampus), they allow us to test the more specific hypothesis that it is specific brain regions, not all brain regions generically, that correlate with behavioral function. Unfortunately, it was not possible to subdivide the neocortex into smaller units (notably frontal and temporal lobe volumes) since the database distinguishes only between primary visual cortex (area V1) and the rest of the neocortex. In some cases, values were missing for some species with performance data (*Pongo*, *Papio*, *Hylobates*, *Lagothrix*) and these species were dropped from the analyses. However, in order to determine how their performance affected the analyses, we have included analyses of gross brain measures (brain, neocortex, non-neocortex and neocortex ratio) over the full species complement in the supplementary information. As brain size scales allometrically with body size, we compared the performance of a model incorporating Log body mass as a control. This method effectively evaluates the independent effects of total size plus the additional effect of relative size for a given body size.

Statistics

Closely related species are likely to share traits by common descent, thus patterns of covariance between related species needs to be taken into account (Harvey & Pagel, 1991). We used Purvis' (1995) primate phylogeny and estimated branch lengths according to Grafen (1989), using the default power of one (no transformation). The appropriateness of this approach has previously been tested on a similar dataset (Deaner et al., 2007).

As the complement of species available for each task and the tasks executed by each species varied (see Table 1), we, therefore, included ability on individual tasks in our analysis. For each task, we calculated Phylogenetically Independent Contrasts (PIC, Felsenstein, 1985) using the PIC command in the R (Ihaka & Gentleman, 1996) APE (Analysis of Phylogenetics and Evolution) package (Paradis, Claude, & Strimmer, 2005). PICs calculate contrasts between tips on a phylogeny, which remove the phylogenetic auto-correlation between related species. By default, the PIC function scales contrasts by their variance, which should remove scaling artifacts (see Garland, Harvey, & Ives, 1992). However, to ensure the contrasts were appropriate, we also checked that they were normally distributed and that there was no correlation between scaled contrasts and their standard deviations. As both of these assumptions were met, we aggregated these task-by-task contrasts into a single database. Models were run which evaluated PICs of ability as a function of PIC's of brain components with task as a factor. MAMs (minimum adequate models) were identified by forward and backward stepwise selection, where all brain measures remaining in the model remained significant.

Second, we used information criteria to select the best performing model. Due to small sample size, we used BIC, Bayesian Information Criterion, (Johnson & Omland, 2004). BIC is appropriate for comparing non-nested models with small sample size (Burnham & Anderson, 2002). We report BIC and model weight to

allow for a comparison of model performance. The relative weight of evidence for each candidate model was calculated using the following equation: $w_i = \exp(-1/2\Delta_i / \sum \exp(-1/2\Delta_i))$, where Δ_i represents the change in BIC between the best model (lowest BIC) and alternative models. Factor weights were estimated by summing weights over all models incorporating each factor. For comparison, we also evaluated nonphylogenetic models, where the ability measure was evaluated as a function of brain measure and task. For nonphylogenetic models, MAMs were identified, and BIC and model weight are reported.

For individual tasks, we used Phylogenetic Generalized Least Squares (PGLS) that allows the degree of phylogenetic autocorrelation to be estimated and incorporated into models (Grafen, 1989; Freckleton, Harvey, & Pagel, 2002). The PGLS approach was executed in R (Ihaka & Gentleman, 1996) using the APE (Analysis of Phylogenetics and Evolution) package (Paradis et al., 2005) and code provided by R.P. Duncan. We used Pagel's (1999) λ to incorporate the phylogenetic covariance. Lambda transforms expected covariance according to a specified value ($\lambda = 1$ assumes phylogenetic dependence in the variation of the trait such that the degree of covariance between two species is directly proportional to the evolutionary distance; $\lambda = 0$ assumes that the trait is independent of shared ancestry). Alternatively, λ can be optimized through an iteration process that identifies the λ associated with the maximum likelihood (see Freckleton et al., 2002). The likelihoods of models where $\lambda = 1$, $\lambda = 0$ and an optimized λ can then be compared to determine which model best fits the data.

We used this approach to evaluate separate tasks according to the following protocol:

- a. First, a PGLS analysis was run with lambda (degree of autocorrelation) set to one.
- b. Second, PGLS models were run with an optimized λ (PGLS.lam).
- c. Third, we ran nonphylogenetic models to obtain likelihoods.
- d. We then compared maximum likelihood between PGLS, PGLS.lam, and nonphylogenetic models. Where there was no significant difference in model fit, the simplest (least parameterized) model was retained.
- e. For the retained model (either PGLS, PGLS.lam, or non-phylogenetic), we report BIC for the model fit for individual brain components. Smaller values represent better fits and differences of >2 in the BIC value between models indicates substantially better fit of the smaller value.
- f. Finally, we present correlation coefficients to give an indication of the variance explained by the different models.

Results

All measures of brain components were individually highly correlated with performance on tasks, both in the phylogenetic and GLM analyses (see Table 2). However, it is unsurprising that there

Table 2
Univariate Results

Model	Factors	Nonphylogenetic					Phylogenetically independent contrasts				
		β	df	F	P	Adj R^2	β	df	F	P	Adj R^2
1	Task		7,56	6.96	<0.001	0.58		7,48	2.64	0.02	0.35
	Log hippocampus	0.37	1,56	37.79	<0.001		0.28	1,48	16.75	<0.001	
2	Task		7,56	6.84	<0.001	0.58		7,48	2.48	0.03	0.31
	Log brain	0.29	1,56	35.18	<0.001		0.19	1,48	12.52	0.001	
3	Task		7,56	6.85	<0.001	0.57		7,48	2.49	0.03	0.31
	Log nonvisual cortex	0.24	1,56	34.91	<0.001		0.18	1,48	12.91	0.001	
4	Task		7,56	6.86	<0.001	0.57		7,48	2.49	0.03	0.31
	Log neocortex	0.25	1,56	34.81	<0.001		0.18	1,48	12.74	0.001	
6	Task		7,56	6.79	<0.001	0.57		7,48	2.54	0.03	0.28
	Log medulla	0.37	1,56	34.06	<0.001		0.24	1,48	10.5	0.002	
7	Task		7,56	6.67	<0.001	0.57		7,48	2.47	0.03	0.29
	Log non-neocortex	0.30	1,56	33.92	<0.001		0.19	1,48	1.11	0.002	
8	Task		7,56	6.77	<0.001	0.56		7,48	2.49	0.03	0.31
	Log cerebellum	0.25	1,56	32.53	<0.001		0.17	1,48	12.26	0.001	
9	Task		7,56	6.7	<0.001	0.56		7,48	2.57	0.03	0.29
	Log visual cortex	0.36	1,56	31.87	<0.001		0.25	1,48	11.29	0.002	
11	Task		7,56	6.81	<0.001	0.50		7,48	2.44	0.03	0.27
	Neo ratio	0.25	1,56	21.24	<0.001		0.21	1,48	9.48	0.003	
7	Task		7,55	6.87	<0.001	0.59		7,47	2.39	0.04	0.29
	Log brain	-0.13	1,55	0.32	0.57		0.16	1,47	0.72	0.4	
	Log body	0.28	1,55	3.06	0.09		0.03	1,47	0.04	0.85	

Note. β represents slope of relationship. Task was included in all models to control for task performance differences and avoid pseudoreplication.

is little variation in the results, given that brain component volumes are strongly correlated with one another. We therefore ran a model comparison exercise using alternative brain measures to tease apart statistically independent effects.

Over all tasks, both the nonphylogenetic and BIC approach identified the same best performing models (i.e., those that differ from the best performing model by a BIC value <2) (see Table 3). All of these models incorporated hippocampus size,

Table 3
Best Models (Both Phylogenetic and Non-Phylogenetic)

Model	Factors	df	β	F	P	Nonphylogenetic (GLM) analyses			Phylogenetic contrasts		
						Adj r^2	BIC	Weight	Adj r^2	BIC	Weight
1	Task	7,54		8.91	<0.001	0.65	-19.72	0.20	0.53	9.11	0.21
	Log hippocampus	1,54	1.17	13.73	<0.001						
	Neocortex ratio	1,54	0.44	12.01	0.001						
	Log brain	1,54	-0.86	8.11	0.006						
2	Task	7,54		8.85	<0.001	0.65	-19.5	0.18	0.53	9.26	0.20
	Log hippocampus	1,54	1.22	13.10	0.001						
	Neocortex ratio	1,54	0.3	12.12	0.001						
	Log non-neocortex	1,54	-0.9	7.91	0.007						
3	Task	7,54		8.91	<0.001	0.60	-19.4	0.17	0.52	9.99	0.14
	Log hippocampus	1,54	1.06	13.92	<0.001						
	Log nonvisual cortex	1,54	-0.74	7.77	0.007						
	Neocortex ratio	1,54	0.48	11.37	0.001						
4	Task	7,54		8.57	<0.001	0.65	-18.8	0.13	0.53	9.30	0.19
	Log hippocampus	1,54	1.42	13.52	0.001						
	Log nonvisual cortex	1,54	1.21	11.43	0.001						
	Log non-neocortex	1,54	-2.3	10.83	0.002						
5	Task	7,54		8.55	<0.001	0.64	-18.28	0.10	0.52	9.94	0.14
	Log hippocampus	1,54	1.39	13.01	0.001						
	Log neocortex	1,54	1.19	10.89	0.002						
	Log non-neocortex	1,54	-2.21	10.29	0.002						

Note. F values are from General Linear Models (GLM) Bayesian Information Criterion (BICs) calculated from generalized linear models. Phylogenetic models use Phylogenetically Independent Contrasts (PICs). See Method for description of BIC and weight.

together with a measure of neocortex volume and a scaling variable (e.g., brain size, medulla, or non-neocortex). Note that these measures control for a negative relationship between cognitive performance and the relative volume of the subcortical brain units. In contrast, hippocampal and (overall and sub-region) neocortex volume were positively associated with performance. The weights of all models incorporating alternative factors were summed and the resulting factor weights for non-phylogenetic/phylogenetic models, in descending order, were: hippocampus 0.93/1.00, neocortex ratio 0.57/0.56, subcortical volume 0.46/0.59, nonvisual cortex 0.38/0.43, total brain volume 0.30/0.23, neocortex volume 0.13/0.15, visual cortex 0.04/0.03, body size 0.05/0.03, medulla 0.03/0.01, and cerebellum 0.01/0.01.

Task-by-Task Analysis

As with overall models, hippocampus volume was the brain measure most consistently correlated with performance (Table 4). However, it must be noted that the correlation coefficients between brain regions often varied rather little. We additionally ran phylogenetic (PGLS, PGLS.lam) and nonphylogenetic models for all tasks. In addition to the brain measures included in the correlation table, we ran models that included both body mass and total brain volume as predictors, and one with log-neocortex and log-subcortical volume as predictors. These models are logically equivalent to using residuals of brain size, as they control for the effect of body size and the effect of total subcortical volume respectively. In no case did the PGLS ($\lambda = 1$) provide the best fit (Figure 1). In some cases, the optimized lambda, and in other cases the nonphylogenetic model, had the highest likelihood. However, as the likelihoods between these latter two models did not differ significantly, we report only the BIC for the nonphylogenetic model as it has the fewest parameters (Table 5). Performance on both delay and detour tasks were best predicted by brain and body mass, whereas for reversal, oddity, and string tasks, there was little difference in the predictive power of different brain components.

Discussion

These results demonstrate that, contrary to Macphail's Null Hypothesis, consistent quantitative species differences in performance on executive function tasks do exist in primates. What is more important is that these differences are a function of both brain volume and architecture. These results add weight to recent claims for a significant

correlation between brain volume and general intelligence (g or IQ) in primates (Deaner et al., 2007) and humans (Andreassen et al., 1993; Reiss, Abrams, Singer, Ross, and Denckla, 1996; Deary & Caryl, 1997; Wickett, Vernon, & Lee, 2000; Postuma et al., 2002; Pennington et al., 2000). Our results are consistent with those of Deaner et al. (2007), in that overall brain size was an important predictor of performance. In addition, however, we were able to identify the additional role of the hippocampus as an important predictor of ability. The high correlation between brain component size, overall brain size, and body size results in univariate tests being poor discriminators between potential causal relationships. Thus, our model selection approach additionally allowed a comparison of the relative predictive ability of different measures. Our best performing models all included a measure of total brain size, neocortex ratio and hippocampus. It is important that in all of the best performing models, brain size, and subcortical volume had *negative* slopes, and thus seem *not* to be crucial for executive function cognition. In effect, they act as a control for subcortical brain components. Thus, our analyses indicate that it is the cortex and hippocampus that are explicitly associated with executive functioning performance in primates. This challenges the results of Deaner et al. (2007), whose univariate analyses are unable to partition out the relative effects of the different components. Although their effort was commendable and enlightening, the aggregate performance values both introduced additional error and reduced the predictive power of their analyses. Finally, body size had a low overall weighting as a parameter in the models. This together with the strong allometric relationship between brain and body size suggests that conventional measures of residual brain size are likely to confound rather than improve relationships between brain size and cognition, especially within closely related taxonomic groups.

Extensive evidence links the hippocampus to learning, both with spatial and nonspatial tasks. In both humans and nonhuman animals, lesions to the hippocampus result in learning impairment, especially where associations between stimuli or novel spatial environments are involved (reviewed by McClelland et al., 1995). This dependence on the hippocampus for forming associations is consistent with the data we present, which show that performance on most of the executive function tasks were significantly associated with hippocampus size, especially for the association and memory tasks. However, even though the relationship with the hippocampus is particularly strong, it cannot of itself explain the very large volumetric differences in brain size that exist between

Table 4
Correlation Coefficients Between Task Performance and Brain Size Measures

Description	Neocortex ratio	Log brain	Log neocortex	Log non-neocortex	Log medulla	Log visual cortex	Log hippocampus	Log cerebellum	Log nonvisual cortex
Delay ($n = 9$)	.601	.698*	.677*	.719*	.683*	.651	.723*	.646	.675*
Detour ($n = 7$)	.544	.876**	.856*	.897**	.888**	.855*	.913**	.855*	.856*
Displace ($n = 6$)	.632	.701	.697	.704	.706	.643	.754	.701	.698
Oddity ($n = 8$)	.592	.575	.594	.536	.572	.630	.494	.605	.593
Reversal ($n = 9$)	.733*	.779*	.783*	.762*	.762*	.750*	.735*	.766*	.783*
Reverse reward ($n = 7$)	.530	.838*	.828*	.852*	.880**	.858*	.884**	.840*	.829*
String ($n = 8$)	.738*	.747*	.760*	.717*	.745*	.724*	.726*	.768*	.762*

* Indicates significance between $0.01 < p < .05$. ** at $p < .01$. Boldface indicates the largest correlation coefficients.

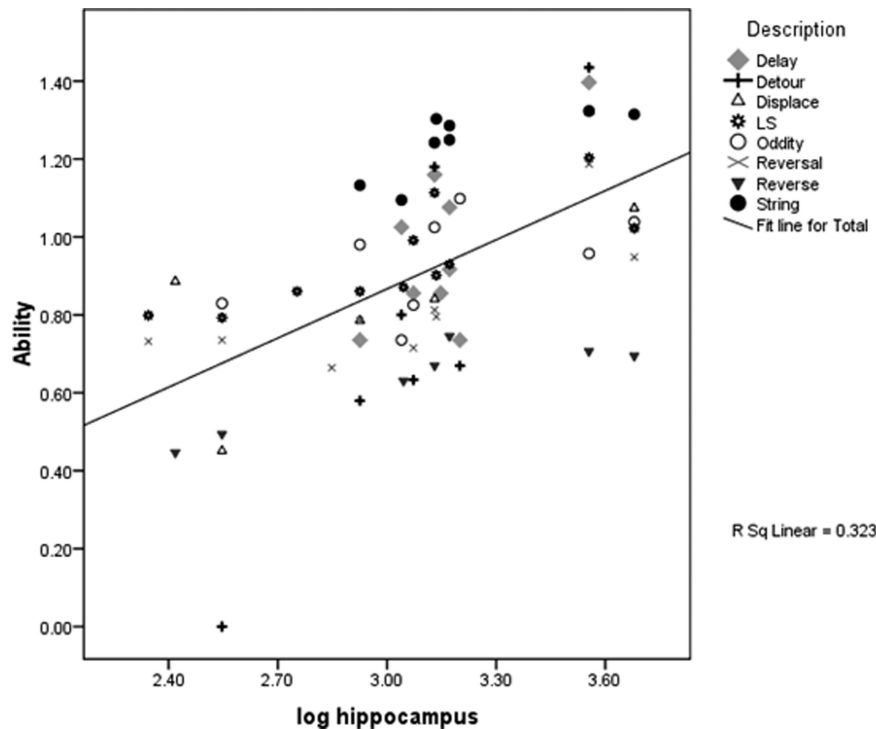


Figure 1. The relationship between task performance and the log of hippocampal volume for individual species on each task.

species. Something else must be central to the demands of executive function cognition, and our results imply that this must be associated with a network of processing units that is distributed throughout much of the neocortex.

How do these results illuminate the domain general versus modularity debate? Fodor's (1993) interpretation of modularity is that domain specific cognition is most relevant for specific inputs, such as the visual system. One proposal has been that the hippocampus is effectively a module for processing conscious information (Muscovitch, 2008). However, the hippocampus is not solely responsible for learning. We know, for example, that the neocortex, and specially its medial temporal and frontal lobes, is involved in the acquisition and consolidation of memory, as well as in discrimination and problem-solving tasks (Sutherland &

McNaughton, 2000; Wiltgen, Brown, Talton, & Silva, 2004). One interpretation is that the hippocampus is implicated in encoding and retrieving information, whereas the frontal lobes are implicated in organizing and interpreting the information coded via the hippocampus. The key is that both systems are necessary for learning cognitive tasks, such as those in the battery we have analyzed. In addition to hippocampal volume, neocortex ratio and non-V1 cortex volume were the next strongest predictors of ability. (Unfortunately, the database conventionally used in analyses of this kind does not partition down the neocortex into finer units beyond separating out primary visual cortex, area V1.) That our best performing models incorporate both hippocampus and relative neocortex size is consistent with the suggestion of an integrated distributed system. This is consistent with the brain having com-

Table 5
Task-by-Task Nonphylogenetic Analyses

Task	Brain	Neocortex	Non-neocortex	Neocortex ratio	Brain + body	Neocortex + rest of brain	Hippocampus	Visual cortex	Nonvisual cortex	Cerebellum	Medulla
Reverse reward	-14.37	-13.96	-14.93	-8.18	-12.51	-13.34	-16.51	-15.2	-14	-14.13	-16.32
Delay	-2.57	-2.07	-3.12	-0.6	-11.28	-1.17	-3.22	-1.52	-2.04	-1.42	-2.22
Reversal	-10.39	-10.53	-9.79	-8.91	-8.43	-8.55	-8.97	-9.41	-10.52	-9.93	-9.8
Oddity	-8.48	-8.75	-7.98	-8.72	-9.28	-7.51	-7.52	-9.32	-8.75	-8.93	-8.45
String	-17.98	-18.36	-17.23	-17.76	-16.24	-17.86	-17.46	-17.41	-18.42	-18.59	-17.94
Learning sets	-16.55	-16.27	-16.59	-12.79	-15.29	-14.13	-18.34	-16.13	-16.26	-16.07	-16.72
Displacement	2.47	2.54	2.42	3.47	2.84	4.09	1.49	3.32	2.53	2.47	2.39
Detour	3.53	4.52	2.33	11.29	-6.57	4.08	1.21	4.54	4.48	4.55	2.9

Note. Numbers are Bayesian Information Criterion (BIC) values. Best fit models are shaded. Bold numbers indicate models whose performance are not different from the best-fit model ($\Delta\text{BIC} < 2.0$).

plementary and connected learning systems (as per McClelland et al., 1995): memories are consolidated in the neocortex, where subsequent information processing and control of behavior can occur, while rapid acquisition of associative patterns (with feedback from the neocortex) occurs in the hippocampus. The two systems reciprocally reinstate information via two-way connections.

An obvious extension of this study is to evaluate the relationship between performance on tasks with regions of interest within the neocortex. As the nature of tasks vary (e.g., spatial, discrimination, and inhibition tasks), we would expect performance to correlate with specific neocortical regions where there is domain specificity for particular tasks. Alternatively, where cognition is distributed we would expect overall neocortex or brain size or connectivity (e.g., white matter) to be a better of performance. Advances in imaging make it possible to partition the neocortex into subregions. Using such indices would undoubtedly provide an advance over the Stephan et al. (1981) gross histological database.

An inevitable criticism of comparative studies is that collating data across studies is bound to introduce error from differences in experimental design (including number of subjects, protocols, and individual experience), brain size measures, and reporting (Healy & Rowe, 2007; MacPhail, 1985). These confounds are to some extent unavoidable, as the experiments span a long period of time, and were conducted at different facilities by different researchers. However, we argue that the effect of confounds are likely to be asymmetric: they would be more likely to obscure relationships rather than produce them. If we found no association between brain size and performance, we would need to be cautious about concluding that brain size is unrelated to cognition because such an outcome could easily be the result of compounded errors. However, we found consistent relationships between brain size and cognitive performance, which suggests that these patterns are robust enough to persist despite the “noise” introduced by collating studies that use different methodologies.

These findings have significant implications for both how and why cognitive functions evolve, at least within the primates. First, it allows us to overcome the inherent conflict at the heart of Macphail’s Null Hypothesis: why should some species need so much bigger brains than others if there are no intrinsic differences in cognitive ability. Second, it suggests that one selection pressure favoring brain enlargement has been executive function competences. Previous studies have demonstrated that, in humans, brain architecture is both highly heritable (Baaré et al., 2001; van Leeuwen et al., 2009) and associated with performance on IQ tests (Colom et al., 2009; Luders, Narr, Thompson, & Toga, 2009). The evidence for both intraspecific and interspecific relationships between brain size and cognitive performance thus justifies the use of brain size as a proxy for intelligence. Additionally, the high heritability estimates for brain architecture indicates that selection, via differences in individual performance, can operate on brain architecture, including overall size.

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